



UNIVERSITI TEKNOLOGI MALAYSIA

TASK 1

NETWORK COMMUNICATION

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Table of content

1.0 Introduction

2.0 Floor plan

2.1 First floor

2.2 Second floor

3.0 facilities

3.1 General purpose lab

3.2 Student lounge

3.3 Hybrid classroom

3.4 video conferencing room

3.5 Embedded lab

3.6 Cisco network lab

3.7 Cafe

3.8 Toilet

4.0 Meeting minute

1.0 Introduction

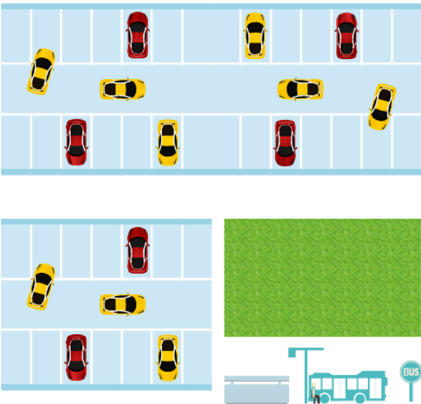
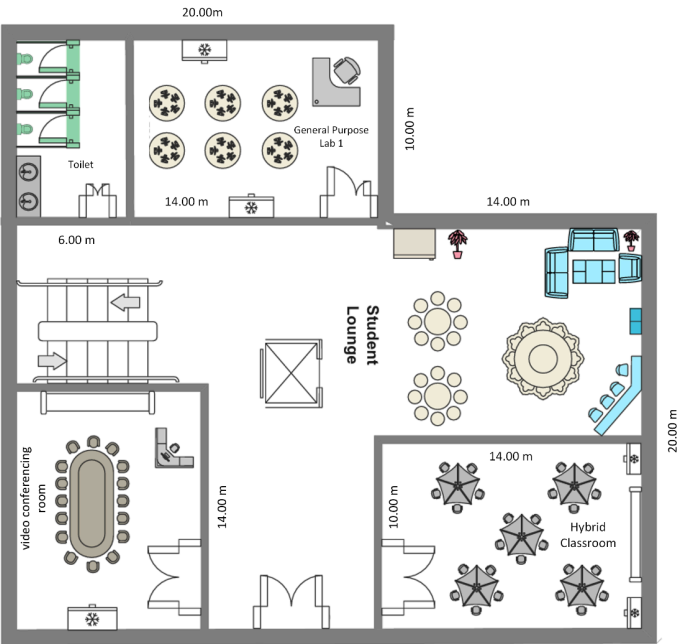
The Faculty of Computing embarked on the ambitious expansion project to comfortably accommodate the anticipated growth in student and staff numbers over the ensuing four-year period. The new two-storey building will add to the current space and house state-of-the-art facilities aligned with the needs of modern education and the Fourth Industrial Revolution on the back of the expected 15% growth in students and academic staff. This will also ensure that this new facility is provided with good, high-speed internet access and all other specialized resources required for hands-on learning, virtual collaboration, and innovation.

The building architecture shall be designed to optimize functionality and access. The first floor will have a general-purpose lab, a student lounge with Wi-Fi to allow students to relax or work, restrooms, stair access, an elevator, a video conferencing room for virtual meetings, and a hybrid classroom designed for maximum learning. The second floor will have the facility of another general-purpose lab, restrooms, stair and elevator access, a café for refreshments, an embedded lab focused on IoT and digital technologies, and a fully equipped Cisco Network Lab with network devices for teaching purposes.

This design meets not only today's demands but also positions FC for future advancements, thus providing a flexible, secure, and cost-effective network infrastructure that aligns with the faculty's educational and technological goals.

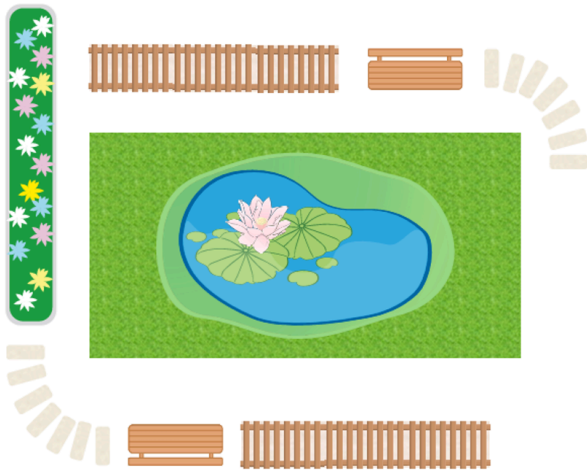
2.0 Floor plan

2.1 First floor

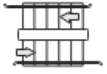


	Door
	Sink
	Table with computer
	Door
	Air conditioner
	Coordinator desk
	Door
	Escalator
	Elevator
	Refrigerator
	Flowerpot
	Sofas
	Big rounded table'
	Rounded table
	Table of 5 person
	Smartboard
	Parking lot
	Field
	Bus stop
	Bench

2.2 Second floor

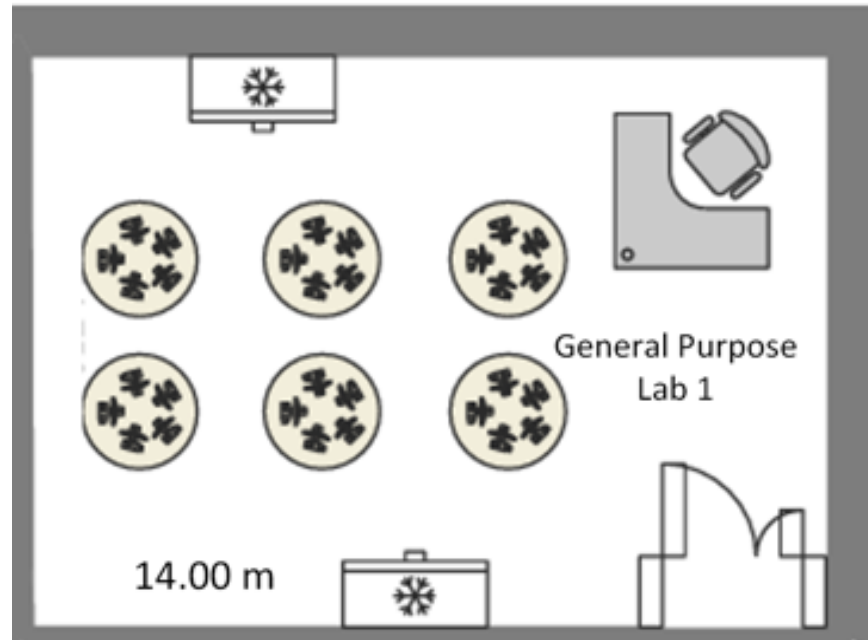


Indicators:

	Cabinet		Sink		Lecturer's table
	Projector Screen		Toilet		
	Air Conditioner		Cafe seatings		Lecturer's table
	Door		Table		
	Cubicle		Staircase		Router
	Cafe table		Plant		Switch
	Cubicle		Door		
	Elevator		Cafe seatings		

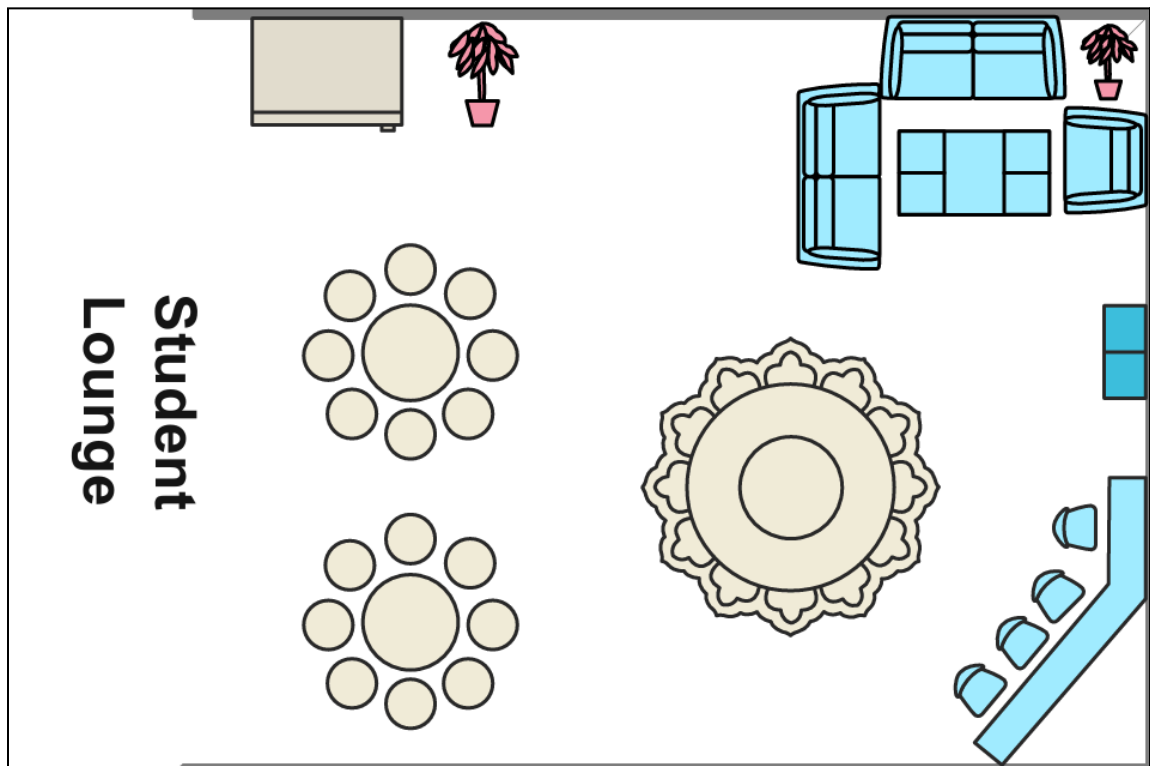
3.0 facilities

3.1 General purpose lab



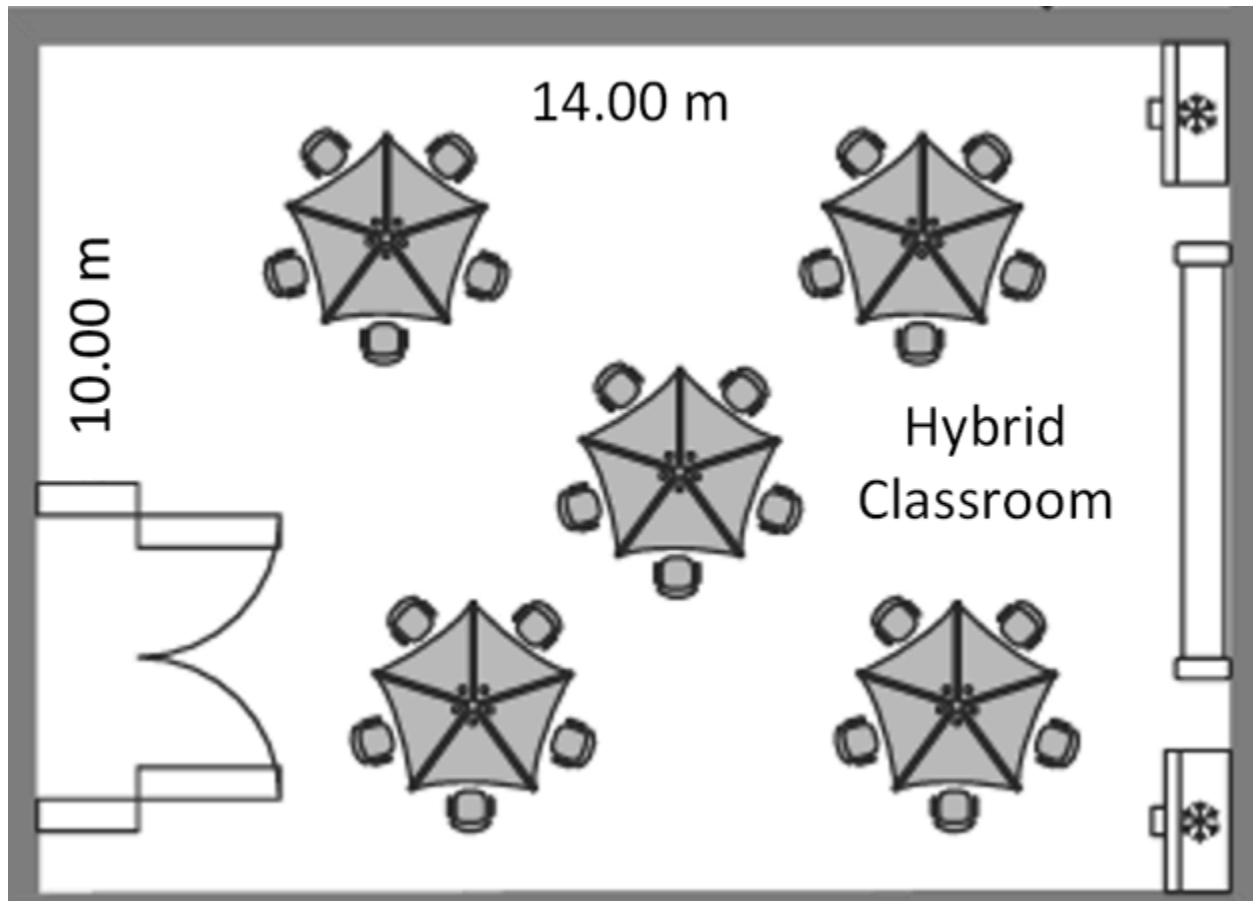
The General Purpose lab is a flexible area room made for accommodating a range of tasks and activities for several subjects. The lab measurement is 14m x 10m, and It could fit 30 students at one time. The lab also provided a pc for every student, so that they can use it at any time. We chose the concept of a circle shaped table because we would like to encourage students to have a discussion among them. This is because, we believe by having a group discussion, it could expand the idea even bigger.

3.2 Student lounge



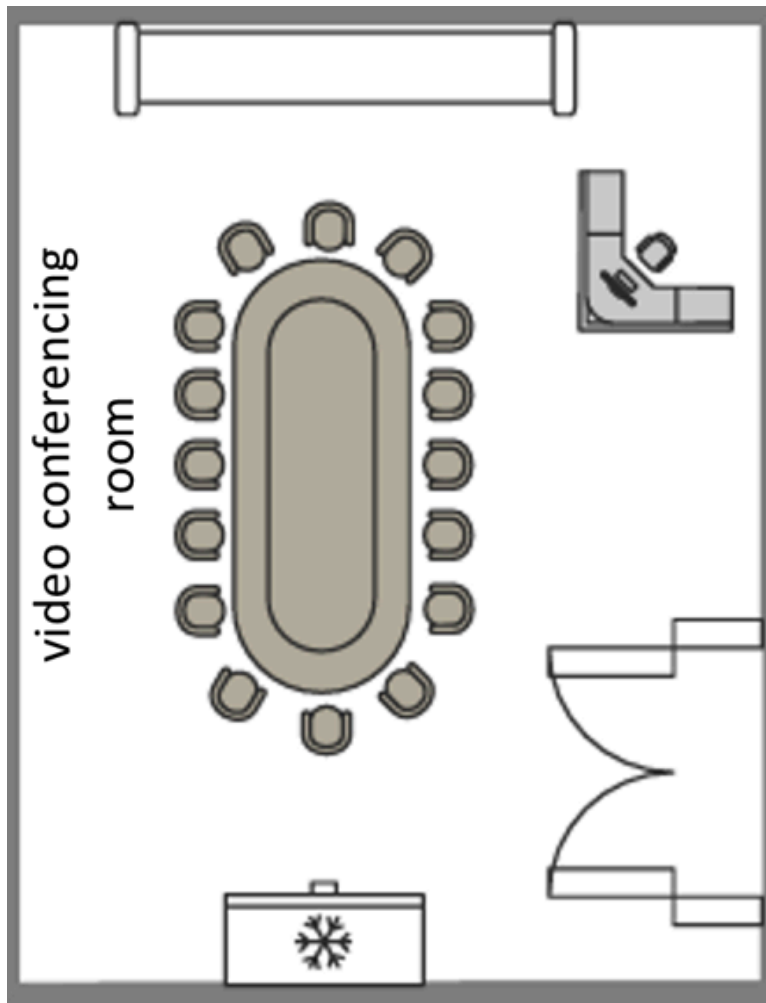
A student lounge is a friendly and adaptable area for students on campus. Students can unwind, mingle, and form relationships in this laid-back setting, which promotes a feeling of community and belonging on campus. Because it has couches and tables, it also offers spaces for both private and group study sessions.

3.3 Hybrid classroom



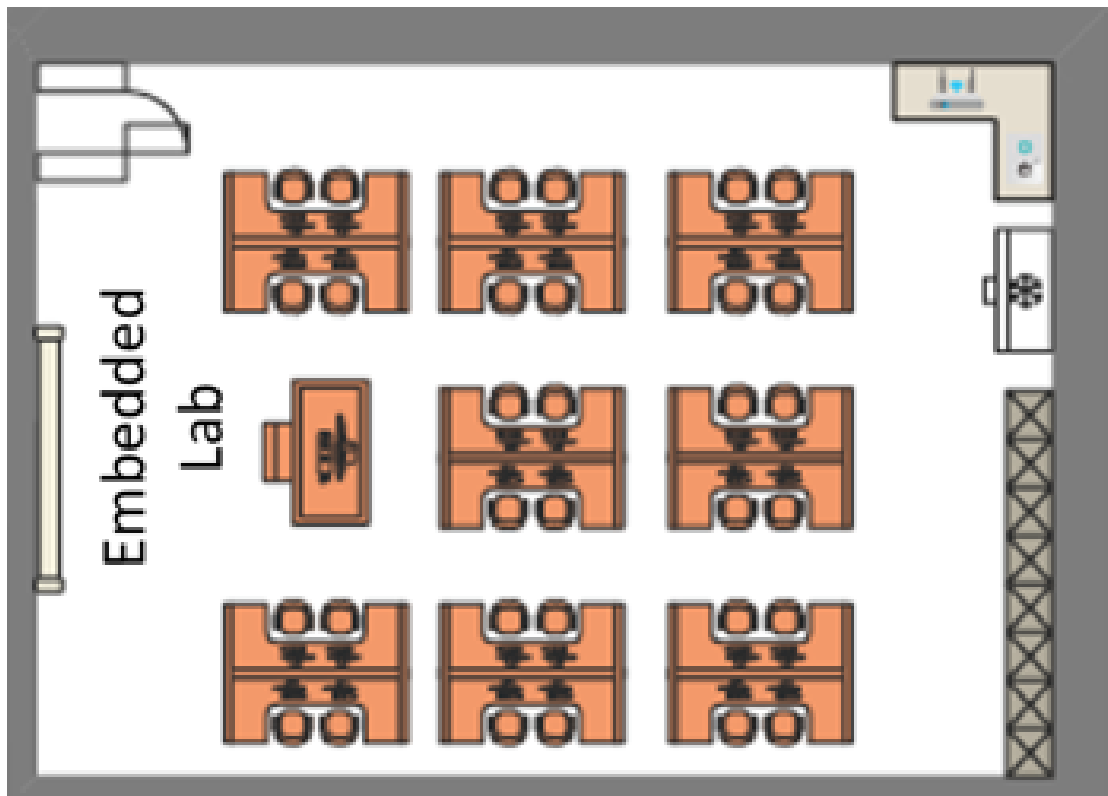
A hybrid room is perfect for flexible learning, teamwork, and meetings since it can hold both in-person and remote participants. In order to create a smooth experience for hybrid learning environments in educational institutions, hybrid rooms allow teachers to instruct both in-person and online learners. The classroom measurement is 14m x 10m, and It could fit 30 students at one time. As a result, teams can collaborate more easily no matter where they are. Aside from that, hybrid spaces are perfect for holding webinars, workshops, and panel discussions, which increase the number of ways that people can participate by allowing them to attend both in-person and virtually.

3.4 video conferencing room



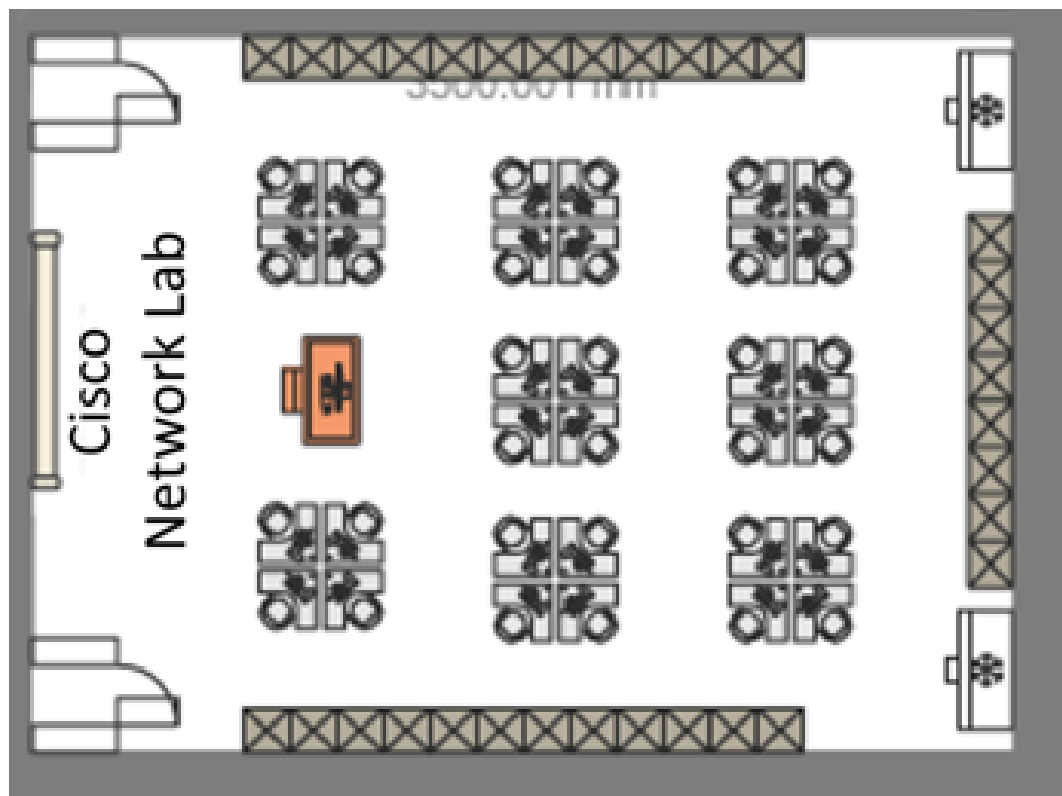
Virtual meetings are supported by a video conferencing room, which makes it possible for individuals in various places to communicate effectively. A video conferencing room with excellent video cameras, microphones, and screens offers crystal-clear audio and visual capabilities to guarantee that participants can see and hear each other clearly. Our video conferencing rooms are soundproofed or equipped with noise-canceling technology to minimize interruptions and safeguard the confidentiality of discussions. As a result, these spaces are ideal for holding online gatherings such as webinars, panel discussions, and Q&A sessions, which enable more extensive audience participation and interaction.

3.5 Embedded lab



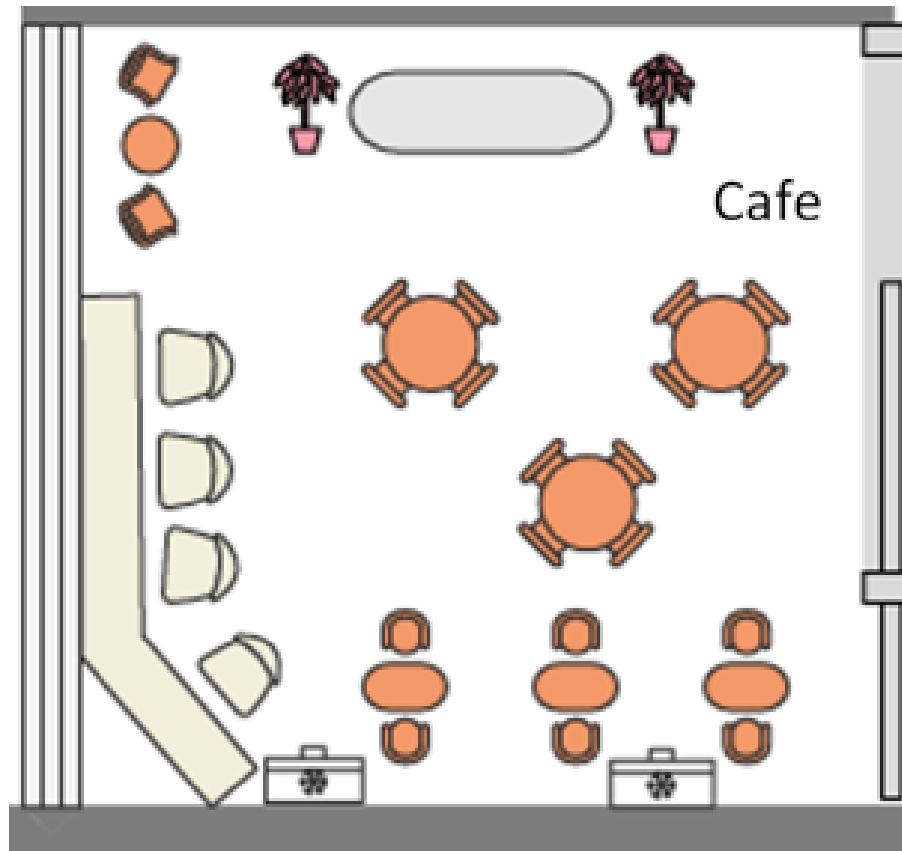
Embedded lab is a specialized laboratory focused on the development, testing, and prototyping of embedded systems. Embedded systems are dedicated computer systems designed for specific tasks or functions within larger devices (like medical equipment, automotive systems, or consumer electronics). In an embedded lab, engineers and technicians work with hardware and software components to design and build these systems, often testing them rigorously to ensure they meet performance, reliability, and safety standards.

3.6 Cisco network lab



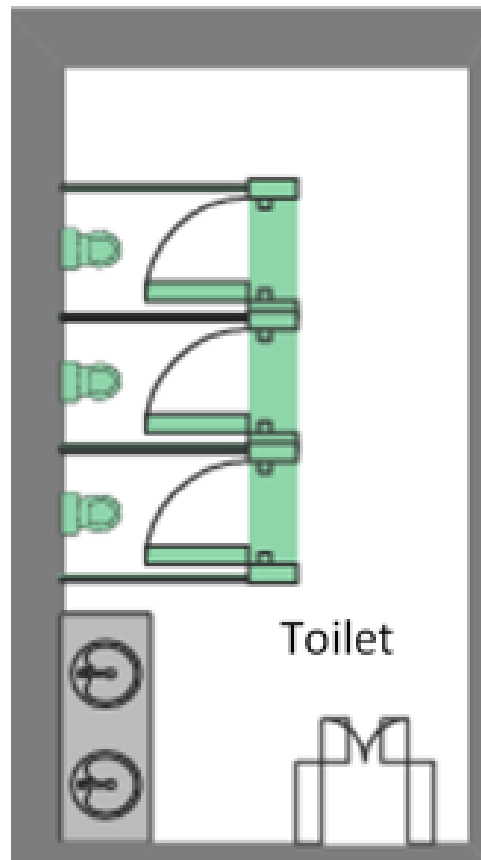
A Cisco network lab is a specialized environment, often used for educational, testing, and research purposes, where Cisco networking equipment (like routers, switches, and firewalls) is set up for hands-on learning and experimentation. It helps students, network engineers, and IT professionals build practical skills in network configuration, troubleshooting, and management, typically aligned with Cisco's certifications.

3.7 Cafe



A cafe is a small establishment where students go to enjoy coffee, tea, and other beverages, often alongside light food options like pastries, sandwiches, and snacks. Unlike restaurants, cafes are typically designed for a relaxed, informal experience where people can sit and socialize, work, or simply enjoy a quiet moment. Many cafes have comfortable seating, free Wi-Fi, and a welcoming atmosphere, making them popular spots for studying, remote work, or meeting friends.

3.8 Toilet



Toilets are typically located in bathrooms or restrooms, in both residential and public settings, and are a fundamental component of sanitation systems. They are engineered to contain waste and flush it into a sewage system or septic tank, ensuring proper hygiene and cleanliness.

4.0 Meeting minute

MEETING MINUTE 1

DATE/TIME	31th of October 2024		
LOCATION	ONLINE - GMEET		
AGENDA	1) Discuss the question given for Task 1 2) Determine the software and tools needed 3) Do a rough sketching of the floor plan 4) Decide what to put on each floors		
Meeting MC	YASMIN BATRISYIA BINTI ZAHIRUDDIN		
ATTENDANCE			
NAME	TIME	REASON FOR ABSENCE	
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI	2:00 PM	-	
AHMAD ADIB ZIKRI BIN A.MAZLAM	2:00 PM	-	
YASMIN BATRISYIA BINTI ZAHIRUDDIN	2:00 PM	-	
SYARIFAH DANIA BINTI SYED ABU BAKAR	2:00 PM	-	
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Discuss the materials given for Task 1	i) Syarifah Dania suggested all of the team members go through the questions and determine the requirements needed to complete Task 1.	All members (31/10)

		ii) All members read the materials given and identify the requirements iii)	
2	Software to use	i) Yasmin Batrisyia and Muhammad Mukhritz did online searching to survey available and easy to use software to design the floor plan. ii) All members agreed to utilize Edraw due to its user-friendly and easy to use interface.	Yasmin Batrisyia (31/10) Muhammad Mukhritz Al Iman (31/10)
3	Floor design and room placement	i) Ahmad Adib Zikri led the rough sketch of the floor design by sharing his Ipad screen on Gmeet. ii) Syarifah Dania suggested placing the hybrid classroom and video conferencing room on the first floor. iii) Muhammad Mukhritz suggested that the student lounge should be on the first floor as well. iv) Yasmin Batrisyia uttered that the General Purpose Lab should be both on the first and second floor. v) Ahmad Adib recommended placing the Cisco and Embedded labs on the second floor. vi) Yasmin Batrisyia suggested installing an elevator for easier access to the second floor.	All members (31/10)

		vii) All members agreed on the placement of each rooms. viii) Rough sketch of the floor plan was completed.	
4	Next meeting	i) All members agreed to conduct another meeting on 2nd of November as everybody is back to campus after Deepavali leave ended. ii) Agenda of the next meeting was decided. iii) Syarifah Dania reminded all members to do some self-learning on how to use Edraw in order to achieve optimal efficiency at the next meeting.	All members (31/10)
5	Meeting ended	After completing the rough sketch of the floor plan and planning the agenda for the next meeting, Ahmad Adib ended the meeting.	Ahmad Adib (31/10)

MEETING MINUTE 2

DATE/TIME	2nd of November 2024
LOCATION	Ar-Rayyan Cafe, KTDI
AGENDA	1) Divide tasks for each members 2) Complete the design of the actual 2D floor plan 3) Finalize the report.

Meeting MC		YASMIN BATRISYIA BINTI ZAHIRUDDIN	
ATTENDANCE			
NAME		TIME	REASON FOR ABSENCE
MUHAMMAD MUKHRITZ AL IMAN BIN MOHD RAFFI		9:00 PM	-
AHMAD ADIB ZIKRI BIN A.MAZLAM		9:00 PM	-
YASMIN BATRISYIA BINTI ZAHIRUDDIN		9:00 PM	-
SYARIFAH DANIA BINTI SYED ABU BAKAR		9:00 PM	-
MINUTES			
NO.	ITEM DISCUSSED	IDEAS/SUGGESTIONS AND PERSON GIVING IT	PERSON IN CHARGE & DATE
1	Task Distribution for each members	i) Muhammad Mukhritz prepared a poll for all members to choose which part that they want to design. ii) Each member chose 2 facilities for them to design on their own. iii) All members decided to design the floor plan completely based on the rubric without adding any additional features first. Additional features would only be decided once we finished designing all rooms and labs mentioned in the requirements.	Muhammad Mukhritz Al Iman (2/11)

2	Estimating all important measurements for the building.	i) Mukhritz and Syarifah Dania did some calculations in order to measure the whole building precisely. ii) L shaped building design was decided by Mukhritz and Syarifah Dania. iii) All members agreed to continue with the measurements and the building shape decided.	Syarifah Dania (2/11) Muhammad Mukhritz Al Iman (2/11)
3	Designing each room and lab according to Task 1 requirements and rubric.	i) Ahmad Adib Zikri designed GPL and the Student Lounge. ii) Syarifah Dania designed the VC room and Hybrid Classroom. iii) Muhammad Mukhritz made the outline for Cisco Lab and General Purpose Lab 1. iv) Yasmin Batrisyia planned the Embedded Lab and General Purpose Lab 2. v) All designs were then combined into an L shaped building with 2 floors. vi) Additional features were added by the members such as a cafe, toilet, car park, bus stop and a recreational park beside the building.	All members (2/11)
4	Floor plan finalization	i) Colors and labeling were added to the floor plan.	All members (2/11)

		ii) Floor plan completed.	
5	Reporting	i) Task distribution for the report was done according to the places designed by each member respectively. ii) Cover page, table of content and Introduction were done by Muhammad Mukhriz. iii) Meeting minutes was done by Syarifah Dania.	All members (2/11)
6	Meeting ended	Yasmin Batrisyia ended the meeting once the floor plan design and report were completed.	Yasmin Batrisyia (2/11)